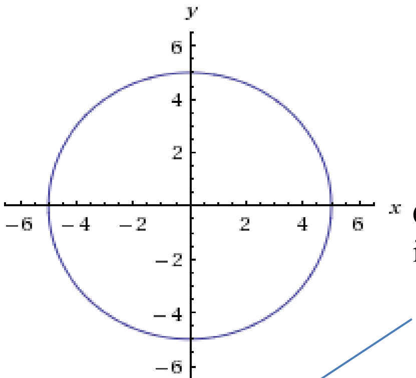
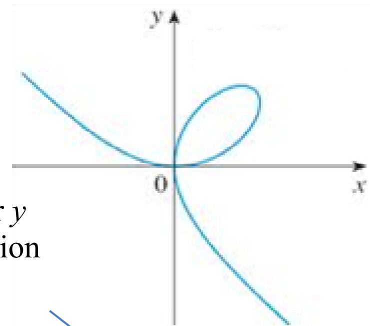
3.8 Implicit Differentiation

<u>Motivation:</u>
Thus far we have mostly dealt with functions where y is a function of x which, by definition, have only one y-value for each x-value in the domain (usually tested for graphically by using a vertical line test). However, some graphs/formulas do not define y as a function of x and we are still interested in being able to find the slopes of tangent lines to such curves.

Consider the following curves (for which y is not a function of x):



Cannot solve for y in a single equation



$x^2 + y^2 = 25$

$x^3 + y^3 = 6xy$

These curves are not functions. Equations like these are called ***implicit equations***. What makes finding a derivative for these curves difficult, is that we defined a derivative as a something we can find for *<u>functions</u>*!

<u>Definitions:</u>

(1) An **implicit equation** is an equation in which one variable is *<u>not</u>* defined in terms of the other variable.

(2) An **explicit equation** is an equation in which one variable is defined in terms of the other variable.

Note: An explicit equation, by definition, is a function.

<u>Example 1:</u> Determine whether each of the following is an implicit or explicit equation.

a) $y = \sin(x) - e^x$

b) $x^2 + y^2 = 25$

Motivation:

Example 2: Consider the equation $x^2 + y^2 = 25$.

a) Solve this equation for y.

b) Find the equation of the tangent line to the curve at the point $(3,4)$.

Example 3: Consider the parabola $x = y^2 - 1$

a) Create a table of values and sketch a picture of this curve.

b) Find the equation of the tangent line to the curve at the point $(8, -3)$.

Question: Why would finding the equation of the tangent line on the graph of $x^3 + y^3 = 6xy$ at the point $(3, 3)$ be difficult?

Definition: Implicit Differentiation is a method for computing the derivative of an implicit equation with respect to the variable x while treating all other variables as unspecified functions of x.

Note: In the book the following unsatisfying statement is made: "The technique of implicit differentiation allows us to find the derivative of an implicitly defined function without ever solving for the function explicitly." This leads to several very important questions:

1. *What does it mean for y to be implicitly a function of x?*
2. *When will the method of implicit differentiation fail?*

Unfortunately, the answers to these questions are beyond the scope of the course. The answers are hinted at in Calculus II and Calculus III. However, one really needs the Implicit Function Theorem which is proven in advanced Real Analysis, to give truly satisfying answers to both of these questions. So, in Calculus I we are in the very unsatisfying position of having to assume that we will only run into questions where implicit differentiation will work.

Example 4: Consider again the equation $x^2 + y^2 = 25$.

a) Find $\frac{dy}{dx}$.

b) Find the equation of the tangent line to the circle given by $x^2 + y^2 = 25$ at the point $(3,4)$.

Example 5: Consider the function $x^3 + y^3 = 6xy$.

a) Find $\frac{dy}{dx}$.

b) Find the equation of the tangent line to the curve given by $x^3 + y^3 = 6xy$ at the point $(3,3)$.

c) What does the derivative evaluate to be at the point $(0,0)$?

Example 6: Find the first derivative of y with respect to x if $\sin(x+y)=y^2\cos(x)$.

Example 7: Suppose that $x^4+y^4=16$.

a) Find $\dfrac{dy}{dx}$.

b) Assuming that $y\neq 0$, find $\dfrac{d^2y}{dx^2}$.

Exploration: Suppose we would like to find derivative formulas for the inverse trigonometric functions. We begin by briefly reviewing the domain and range of these functions.

Function	Domain	Range
$f(x)=\sin^{-1}(x)$	$x\in[-1,1]$	$y\in\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$
$f(x)=\cos^{-1}(x)$	$x\in[-1,1]$	$y\in[0,\pi]$
$f(x)=\tan^{-1}(x)$	$x\in(-\infty,\infty)$	$y\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$
$f(x)=\cot^{-1}(x)$	$x\in(-\infty,\infty)$	$y\in(0,\pi)$
$f(x)=\sec^{-1}(x)$	$x\in(-\infty,-1]\cup[1,\infty)$	$y\in\left[0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right]$
$f(x)=\csc^{-1}(x)$	$x\in(-\infty,-1]\cup[1,\infty)$	$y\in\left[-\frac{\pi}{2},0\right)\cup\left(0,\frac{\pi}{2}\right]$

Note: The notation does NOT mean raised to the negative one power. The negative one indicates that the function is an inverse function. To avoid having to use the negative one, you may also see inverse trig functions written as $\arcsin(x)$, $\arctan(x)$ *, etc.*

Example 8: Find $\frac{d}{dx}\left[\sec^{-1}(x)\right]$.

Example 9: Find formulas for the first derivative of $\tan^{-1}(x)$ and $\cot^{-1}(x)$.

Note: Similarly, we could find formulas for the inverse cosine and the inverse cosecant. The following table summarizes all the derivative formulas of the inverse trig functions.

Derivatives of Inverse Trigonometric Functions:

$\frac{d}{dx}\left[\sin^{-1}(x)\right]=\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}\left[\cos^{-1}(x)\right]=\frac{-1}{\sqrt{1-x^2}}$
$\frac{d}{dx}\left[\tan^{-1}(x)\right]=\frac{1}{1+x^2}$	$\frac{d}{dx}\left[\cot^{-1}(x)\right]=\frac{-1}{1+x^2}$
$\frac{d}{dx}\left[\sec^{-1}(x)\right]=\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$\frac{d}{dx}\left[\csc^{-1}(x)\right]=\frac{-1}{\lvert x\rvert\sqrt{x^2-1}}$

Example 10: Find the first derivative of $y=\frac{1}{\cos^{-1}(x)}$.

Example 11: Find $\frac{dy}{dx}$ if $y=\tan^{-1}\left(\sqrt{x}\right)$.